

Housing Wealth Over the Life Cycle

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Abstract

This chapter summarizes facts and economic theory regarding the accumulation and disposal of housing debt and equity over consumers' life cycles. After reviewing some basic statistics concerning housing wealth in the United States, I turn to the growing body of research concerning the way that homeownership affects households' ability to allocate resources to the dates and states of the world in which they are most needed. Despite decades of marketing efforts, financial products, such as equity sharing and home price index trading, meant to undo certain allocative effects of homeownership have yet to become popular. A striking trend in North America is diminished household headship and ownership among younger people.

1 Housing Wealth Over the Life Cycle: US Statistics

Data from surveys of US consumers highlight the importance of housing as both an asset and a liability. Khater et al. (2024), summarizing U.S. Consumer Expenditure Survey panel data, find that between 1984 and 2023, rent grew fairly steadily as a share of renters' total expenditures, from roughly 18% to 26%. By contrast, owners collectively spent a relatively constant share of expenditures, roughly 15% over time over that period. Mortgage expenditure shares notably fell, not surprisingly given falling interest rates and most owners' option to fix rates, offset by rising property taxes and other shelter expenditures such as maintenance.¹

The greater housing expenditure for renters than owners found in the CEX is consistent with the empirical consensus that the wealth elasticity of housing demand is less than one (that is, the fraction of wealth spent on housing falls with wealth).² In the 2000 IPUMS census data, I find an elasticity of rent paid with respect to social security income (intended as a proxy for lifetime income) of just .3 among retirees living alone who rent housing aged between 65 and 76. In the 2020-2024 5-year ACS, that value rises to .47.³ Rent relative to social security income is higher in larger and more expensive metropolitan areas, an interesting counterpoint to the finding in Survey data also provide strong support for the

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¹The opportunity cost of equity and anticipated capital gains, part of true "user cost" (see, e.g. Poterba (1984)), are not included in owners' expenditure shares.

²See, e.g. Carliner (1973).

³These results are conditional only on metropolitan area dummies and the year for the 5-year ACS.

nearly experimental result of Hanushek and J. M. Quigley (1980) that the price elasticity of demand for housing is less than one (that is, spending on housing rises as the price of housing rises). Despite similar incomes, renters in central cities in the CEX spend 22% of income on shelter, as opposed to 13% among renters in rural areas.⁴

Tabulations of 2004 Survey of Consumer Finances data in Table ?? show that for most homeowners of every age group, housing represents a majority of assets.⁵ For most owners under 50, the value of the home is greater than net worth. The ratio of home value to assets and net worth is declining with age and income. These patterns continued to hold in the 2016 SCF, per National Association of Home Builders (2018).

Figure ?? plots the rate of mortgage indebtedness (a 0-1 indicator) for homeowners by age of household heading homeowners in the 2000 US Census 5% sample and the 2020-2024 5-year ACS (both from Ruggles et al. (2025)). In both samples, not surprisingly, the presence of mortgage debt is very high (over 80%) among mid-20s homeowners, and declines with age well into retirement. Somewhat surprisingly in light of recent increases in prices, mortgage debt has grown less common among younger homeowners and more common among older homeowners. The overall incidence among owners has fallen from 66% in 2000 to 59% in the 2020-2024 ACS.

Figure ?? shows that throughout life, married couples (with spouse present) are more likely to own than rent housing, and far likelier than unmarried individuals.⁶ A substantial portion of the increase in homeownership among household heads in their 20s and 30s comes from increasing marriage rates. Likewise, much of the decline in homeownership among household heads late in life can be explained by an increasing prevalence of widowhood. This is consistent with the finding of Venti and Wise (2000) and others that death of a spouse is among the most important triggers of exit from homeownership among the elderly. Coile and Milligan (2009) use HRS/AHEAD data to show that the downward trend in homeownership in old age accelerates into the 90s, with ill health of individuals joining death of a spouse as a critical cause. They find an ownership rate among individuals of approximately 50% at age 90, but approximately 35% at age 93.

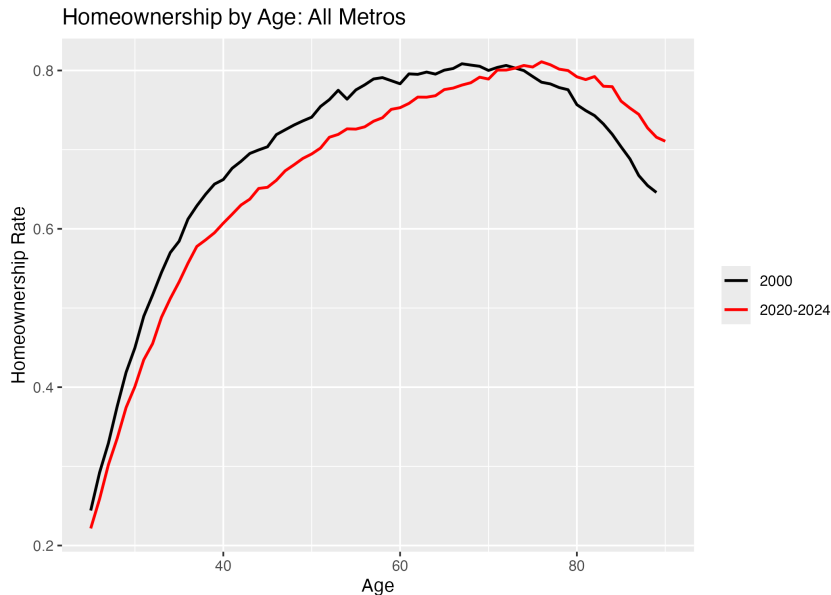
Making use of the patterns depicted in Figure ??, **FisherGervais** show that homeownership rates declined sharply among younger households between 1980 and 2007, attributing this in large part to increasing age of first marriage over time. Increasing overall homeownership can be attributed largely to an aging population and increasing homeownership among the elderly. These patterns have persisted. Per 2000 Census data from Ruggles et al. (2025), among respondents aged 75 and over, in 2000, 69.8% were household heads or spouses of heads and owned their own residence. In the 2024 ACS, it was 72.9%. Among Americans 24-35, in 2000, 46% were household heads or spouses of and owned their home. In 2024, that figure had dropped to 34.9%. This reflects both a reduction in household headship rates and the conditional probability of ownership. While rising housing costs are a global phenomenon, *No Home for The Young? Stylised Facts and Policy Challenges*¹ By Boris Cournède and Marissa Plouin (OECD) shows that between 2007 and 2019, the share of in-

⁴An intriguing result of Davis and Ortalo-Magné (2007) is that despite these low income and price elasticities, the distribution of expenditure shares among renters is very close to constant across time and US metropolitan areas.

⁵These tabulations, unpublished elsewhere were performed by Todd Sinai and Moises Yi.

⁶Those deemed married, spouse absent are dropped.

Figure 1: Homeownership rates by age, 2000 Census and 2020-2024 ACS



dividuals aged 20-29 living with parents was roughly constant overall for OECD countries, but rose dramatically in the U.S. and Canada.

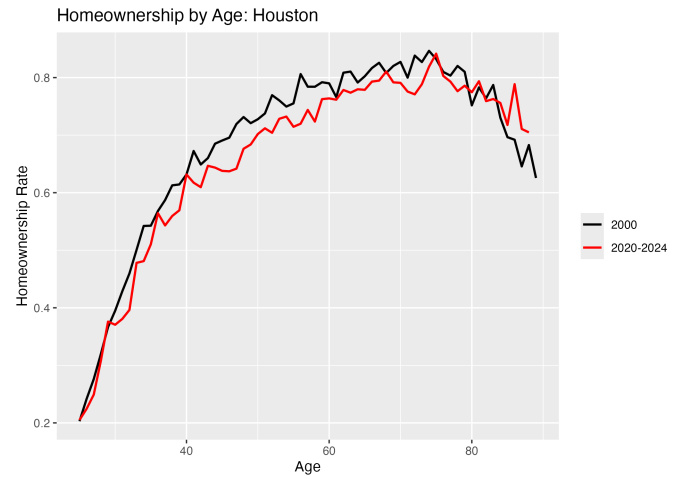
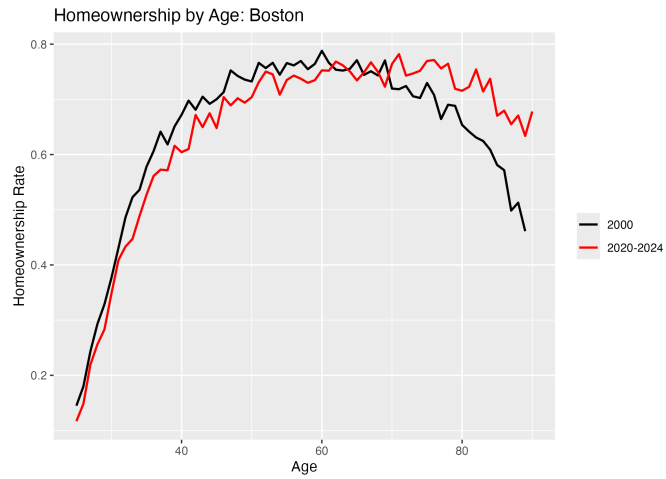
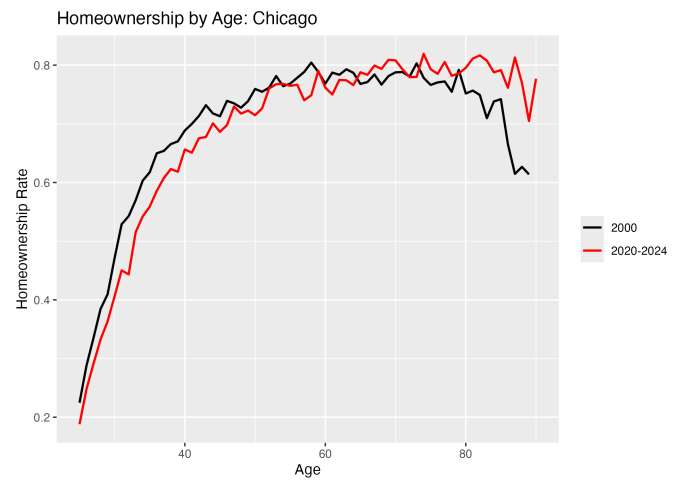
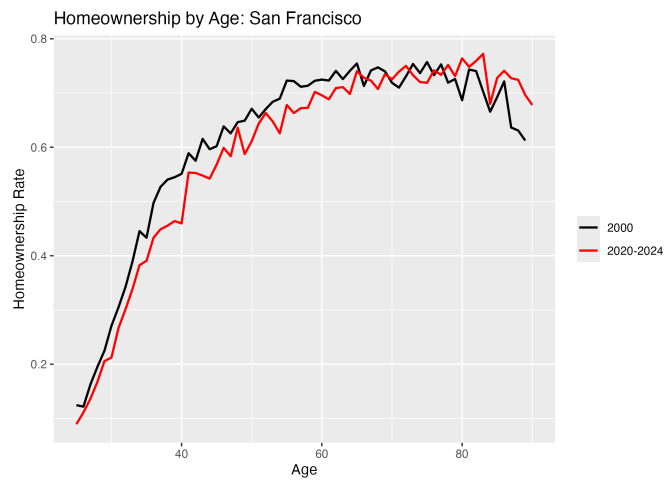
Figure 1 plots homeownership rates among household heads by age for the 2000 Census 5% sample and the 2020-2024 ACS as tabulated by Ruggles et al. (2025). While overall homeownership has been relatively constant, we see large declines at younger ages (even conditional on household headship) and even more persistent homeownership among seniors. Notably, the figures are not notably different in the difference-in-differences among age groups comparing across metropolitan areas. Figure ?? plots the same figure as Figure 1 but separately for notoriously inelastic and increasingly expensive San Francisco, versus Chicago, which has seen more muted price growth (verify).

Among US households, the median and modal number of residential properties owned is one. Aizcorbe, Kennickell, and Moore (2003) find that 67% of households owned housing in 2001, but only 12.8% owned a second home or rental housing.

The unconditional mean income of owners in the 2020-2024 American Community Survey was \$101,988, almost double the mean income of renters (\$57,843). In the 2000 Census (IPUMS microdata), the median homeowner household income was \$49,600, as opposed to the median renter income of \$26,800. This difference in incomes by ownership is consistent with a number of prominent explanations, such as: the US tax code is increasingly favorable to homeownership as income and wealth rise, the normality and age profile of demand for detached homes combined with a natural tendency for detached units to be owner occupied,⁷ and a greater prevalence of credit constraints among those with lower income and wealth.

Consistent with declining tax benefits to homeownership in the U.S. due to elimination of state and local tax deductions as well as reduced mortgage deductibility benefits due to falling rates, rising standard deductions, and caps on mortgage interest deductions, the relationship between homeownership and income weakened slightly over time, with the re-

⁷See, e.g. Glaeser and Shapiro (2003).



gression coefficient on log income (with an indicator for homeownership on the left hand side) falling from .099 in 2000 Census data (taken from Ruggles et al. (2025)) to .084 in 2020-2024 ACS data (taken from Ruggles et al. (2025)). Not surprisingly given the positive geographic cross-sectional relationships among education, home prices, income, and regulations (Davidoff (2016)), these coefficients rise to .113 for 2000 and .096 for 2020-2024, in the presence of controls for metropolitan area fixed effects.⁸

2 Housing wealth in the life cycle: theoretical problems

The institution of homeownership presents a puzzle. Why do most households own exactly 100% of a single home, rather than some smaller share of a diversified bundle of homes? A variety of costs come with full ownership of a particular home, some of which are described below.

2.1 Imposition of mobility costs

For a given household, choosing to own rather than rent housing makes moving costlier. Finding a new occupant for a home requires maintenance and search effort independent of tenure. However, brokerage and transfer fees are significantly larger for owner units, and more marketing effort is presumably required when the next occupant is likelier to be an owner than a renter, and hence expect a longer stay in the home. Purchasing a unit suited for owner occupancy will thus render mobility costly, even if lifetime turnover costs are lower for owner units.

Between the effects of *ex ante* selection and *ex post* costs, it is clear empirically that owners are less mobile than renters. Schachter and Kuenzi (2002) show that in the 1996 Current Population Survey, the distribution of length of time in current residence of owners first order dominates that of renters. The same is true in the 2000 US census. Chetty and Szeidl (2007) find that in response to unemployment shocks, renters are more likely to move than owners, both in levels and relative to baseline mobility. Also, homeowners decrease non-housing consumption more in response to unemployment than do renters. To the extent that housing and other goods are poor substitutes, ownership may thus exacerbate the expected utility costs of the volatility of wealth. In the 2000 US Census, conditional on income, the number of bedrooms in respondents' units declines significantly more rapidly with age among renters than owners, consistent with a greater cost of downsizing for elderly owners.

Whether the existence of demand for changing units makes owning or renting housing relatively advantageous is an open question. The answer will depend in large part on the relative magnitudes of: (a) moral hazard costs of rapid turnover and under-maintenance that arise because tenants do not pay on the margin for landlords' leasing and marketing costs upon lease termination, versus (b) the expected extra financial costs of moving imposed by owner units. The cost (a) is a common rationalization of the institution of homeownership

⁸The full regression specification is a linear probability OLS regression of log income on age dummies and marital status dummies, with the sample selected on household heads ages 25 and older.

through the mechanism of depreciation and maintenance.⁹ The magnitude of cost (b) will depend in part on households ability to substitute consumption between housing and other goods and between current housing and future expenditures. To the extent that homeowners purchase homes that are particularly well-suited to their idiosyncratic needs or attachment to the home grows over time in a beneficial way (for example, through customization) cost (b) will be smaller.

2.2 Excessive investment in particular homes

J. V. Henderson and Ioannides (1983) observe that there is a welfare cost to homeownership if investment demand for a consumer's optimally chosen home is less than consumption demand for that home (in the opposite case, one can invest in housing assets beyond one's own home). There are two obvious ways in which realistic market imperfections will render investment demand relatively weak. First, homeowners may need to hold home equity greater than the amount of total savings they would otherwise choose. Second, homeowners may allocate too much portfolio weight to housing in general and the price risk of their own home in particular.

2.2.1 Too much saving early and late in life

Income for an individual or couple is typically hump shaped over the life cycle. If a flat trajectory of expenditures is desirable, then transfers from middle age to earlier and later years are also desirable. As emphasized by Artle and Varaiya (1978), homeownership typically involves saving for a down payment in youth, and retaining home equity through much of retirement. Both of these characteristics of homeownership interfere with transferring resources from middle age to the early and late years of low earnings.

Both casual and formal empirical analysis demonstrate that down payment constraints are undesirable from the perspective of younger households. A first piece of evidence is the popularity of very high loan to value mortgages in recent years, in a period in which lenders appeared to underestimate the probability and severity of default on such loans. Many such loans featured high payments in the later years of the loan's life, suggesting a willingness to pay for the convenience of avoiding a large down payment. That housing prices increased sharply in response to expansion of mortgage credit also suggests that credit constraints are binding, as in the model of Ortalo-Magné and Rady (2006).

Using microdata covering years in which high loan to value ratios were unavailable or extremely costly, Englehardt (1996) shows that consumers spend less money on food in the years leading up to a first purchase of housing than they do in subsequent years. This again suggests that there are significant liquidity constraints associated with down payments.

The elderly retain a strikingly high quantity of home equity, a fact that has been documented by numerous authors such as Venti and Wise (2000). As shown in figure ??, home ownership rates remain high very late into life, and mortgage debt is decreasing, not increasing in age. As reported in Davidoff (forthcoming), in the 2004 wave of the HRS/AHEAD panel, approximately 12% of retired homeowners over 62 owe any mortgage debt, and among

⁹See, e.g. Shelton (1968).

these, the median debt to value is just 33%. Between 1998 and 2004, despite very large increases in home values, only approximately 10% of retired homeowners increased mortgage debt. Almost all high wealth households have substantial home equity. In the top total wealth quintile of the retired and over 62 sample, 84% held home equity greater than \$100,000. To the extent that retirees wish to spend down their housing wealth before death, it thus appears that a combination of the institution of homeownership and supply problems in the markets for residential sale-leasebacks and mortgages for the elderly stand in the way.

That there is a desire to spend down financial resources before death is not at all obvious, as demonstrated by ambiguous findings on the overall savings rate of the elderly, e.g. Hurd (1992), Horioka (1996) and Weil (1994). Even without a desire to bequeath wealth to heirs, a low level of consumption in retirement may be justified by a need for precautionary savings. Precautionary savings might arise for a variety of reasons, such as a stochastic need for long-term care and uncertain longevity, combined with imperfect insurance markets. Indeed, Davidoff (2009) shows that the welfare benefits of each one of home equity extraction products, annuities, and long-term care insurance depend on the presence or absence of the others.

A critical input to the retiree's financial planning problem is the question of how costly in financial and utility terms it would be to move out of one's home. If it is easy to move out of the home, then a reverse mortgage is presumably less desirable, and the need to maintain a buffer stock of other assets is lessened. Survey evidence in Bayer and Harper (2000) suggests that moves are quite costly. Engelhardt shows that random increases in social security income and support for home health care provision increase homeownership. Coile and Milligan (2009) show that ill health is a crucial determinant of exit from homeownership. A reasonable framework for analyzing homeownership in old age would thus pose a trade-off between the liquidity and potential financial gain to selling a home versus the health-dependent utility from remaining in place. Calibration of such a model for a given household to provide normative advice on savings and optimal move date would stretch economists' tools.

2.2.2 Excessive portfolio weight on a particular home

The possibility that homeownership might lead to excessive portfolio weight on housing has been the subject of growing interest over the last decade (see, e.g. Flavin and Yamashita (2002) and Brueckner (1997)). Setting aside moral hazard considerations described above, it is hard to believe that massive exposure to the considerable idiosyncratic risk embodied in a particular house would be part of an optimal allocation. Establishing an optimal portfolio share for a long position in some market's housing prices in general, however, is extremely difficult. I outline some of the difficulties in determining this optimal share below.

To determine the optimal portfolio exposure to home prices, one must be able to determine whether an investor is better or worse off if home prices rise. The sign of the welfare effect of a price increase depends in part on the extent to which elderly homeowners plan to use home equity to finance retirement. If homeowners simply purchased a home when young, and then used the proceeds to finance retirement, and if rents were uncorrelated with the price of a home, then increasing price variability would reduce portfolio demand for housing, and increasing the size of the home would make the portfolio riskier.

At the other extreme, if housing demand is constant forever and prices are perfectly correlated with rents, then owning a home is a perfect hedge against expenditure risk, and increases in price variability would increase the value of homeownership relative to renting. Given the ability to substitute between housing and other consumption in an otherwise steady state world, a familiar result is that a one-time change to housing prices must be weakly welfare increasing for a homeowner.

Not just hedging demand, but also real options associated with timing of moves, investment, default, and mortgage refinancing may make price volatility more tolerable to owners at any stage of the life cycle. In the recent boom, it seems highly likely that demand and prices were increased by owners and investors in housing who viewed high loan-to-value loans as essentially call options on volatile local prices.

The fact that many homeowners die without cashing out their home equity may be seen as evidence that capital gains risk is of limited importance. However, if housing is held as a buffer for unlikely events, price variation may impose considerable costs as the states in which the home is sold and prices are low may feature particularly high marginal utility.

Just as volatility of prices may increase or decrease the benefits of purchasing more or less housing, volatility of rents has ambiguous effects on the welfare of renters. If housing and other goods are highly substitutable within and between periods, then variability of rents may increase the welfare of renters (this is obvious in the case where housing and other goods are perfect substitutes). In this case, there may be no hedging demand for homeownership. That demand for housing is sufficiently flexible to rule out a motivation to hedge future expenditures runs counter to intuition, and seems unlikely given price elasticity estimates below one.

Among older households, Skinner (1996) observes that it is difficult to distinguish the relative importance of bequest motives, precautionary considerations, and optimal risk averse exercise of an option to sell in the retention of home equity in old age. Indeed the analysis and caveats of Evans, V. Henderson, and Hobson (Forthcoming) suggest that just the relationship between optimal date of sale and price volatility is almost impossible to sign. Some evidence that capital gains risk matters, supportive of a precautionary motive, is provided by Banks et al. (2007) and Sinai and Souleles (2005). Both find that older households tend to sell out of high price volatility environments in old age. The authors' favored interpretation of this evidence is that hedging considerations dominate in youth, so that volatility may boost portfolio demand for housing, but capital gains risk dominates in old age so that volatility decreases portfolio demand for housing late in the life cycle.

An easy test of the notion that homeowners are exposed to too much average price risk comes from the fact that consumers who recognize that they have excessive exposure to housing in their portfolios would seek to avoid exposing themselves to further home price risk with their other investment choices. To my knowledge, there is no study of the cross sectional relationship between (a) the correlation of assets with real estate returns and (b) their relative prices. Such a study would face the challenge that real estate has historically had a weak correlation with financial asset prices, but crashed with the stock market in the great depression and in the recent downturn.

Active investors may or may not demand compensation for exposure to real estate price risk, but these investors are unusual. Findings from the Survey of Consumer finances reported by Aizcorbe, Kennickell, and Moore (2003) show that many homeowners are not

active investors in other markets. As of 2001, approximately 50% of households owned no equities, and many of those who did were passive investors in managed funds. Even unpriced exposure to systematic risk in housing markets does not imply that homeowners have an optimal share of their portfolio in their own particular home. Home prices are imperfectly correlated across markets, so exposure to national or international housing markets is different from exposure to individual market risk. Moreover, home price appreciation varies considerably from neighborhood to neighborhood within markets, and even from home to home within neighborhoods.

3 Minimizing Investment Constraints

A variety of products have been put forward that may help young and old homeowners reduce home equity and to help all owners reduce exposure to systematic and idiosyncratic price risk. An important open question is whether these products have proved unpopular because contracting issues pushes prices away from actuarially fair, or instead because high savings and large housing shares in the portfolio are optimal even with fair pricing.

The obvious way to reduce savings while owning a home is to take on a mortgage. The cost of increasing loan-to-value ratios has varied considerably over time. Recent experience demonstrates that there may be no interest rate at which 100% loan-to-value is profitable to lenders. From lenders' perspective, both refinancing and default are crucial to pricing and profitability, but the optimal exercise of these options for borrowers are difficult to solve even for economists. The recent implosion of the mortgage industry and microeconomic evidence makes clear that we do not have a satisfactory model of default and prepayment, even conditional on complete histories of prices and interest rates.¹⁰ Credit availability to younger homeowners hoping to minimize savings while retaining the hedging, contracting, or psychological benefits of homeownership is thus likely to remain cyclical.

From the discussion above, it is not clear that the elderly wish to reduce savings in the form of home equity or otherwise. The obvious products that would allow the elderly to extract equity without moving out of their homes are sale-leasebacks and reverse mortgages. The latter are somewhat more popular than the former, but even demand for reverse mortgages has been weak, with less than one percent of older homeowners owing reverse mortgage debt. Effective interest rates for reverse mortgages are quite high, with origination fees currently up to 2% of home value and historically considerably higher. Reverse mortgages are likely to feature both adverse selection (witness Mme Jean Calment who lived a record 122 years and had the French version of a reverse mortgage) and moral hazard both on the timing of sales (once a property is worth less than the loan amount, the mortgagor has no incentive to move because the loans are non-recourse) and on maintenance,¹¹ it is not obvious that reasonable loan-to-value ratios can be attained at profitable and marketable rates.

Shiller (1993) and Caplin et al. (1997) have proposed mechanisms whereby homeowners can reduce exposure to variation in home prices. The Chicago Mercantile Exchange now allows investors to take long or short positions on changes in regional average home price

¹⁰Deng, J. Quigley, and Van Order (2000) emphasize the heterogeneity of borrower behavior across households and environments.

¹¹see, e.g. Miceli and Sirmans (1994)

indexes, calculated based on a repeated sale methodology. Several companies have devised mechanisms whereby outside investors can be given equity, rather than debt, in the home. For the latter type of contract, default risk should be lower than with debt contracts, but investment and market timing of sale may lead to problems of moral hazard. Trades on the Case-Shiller CME index avoid homeowner agency problems but do not provide insurance against individual home price risk. Given the complicated effects of home price changes on utility described above, it is not entirely surprising that there has been very limited demand for either type of product.

Summarizing, it is difficult to provide households with easily understood normative advice on major life cycle housing decisions, and particularly difficult to characterize the optimal portfolio share for housing. The conventions of hump shaped life cycle ownership rates, high homeownership rates, mortgage financing, and large unhedged positions in individual homes will be with us for the foreseeable future.